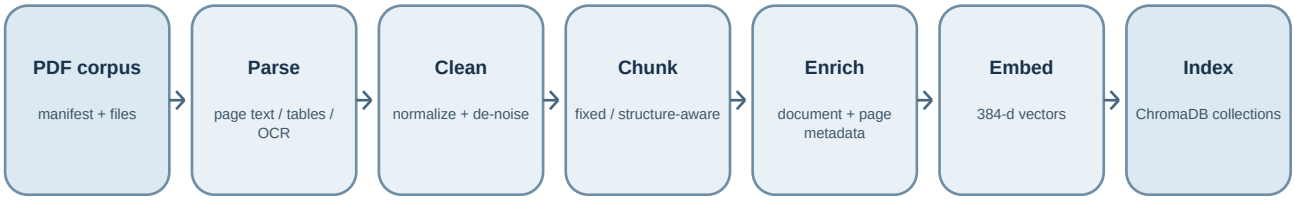
PolicyLens

Ingestion Pipeline for Regulatory Circulars and Internal Policy PDFs

Case Study 2 | Technical Report



Operational controls: source hashing • retry handling • structured logs • replace superseded source versions

Objective. Build a reliable document-ingestion and retrieval foundation for a non-banking lending company's compliance team. The system converts regulatory circulars and policy PDFs into searchable, page-linked passages. Answer generation is deliberately out of scope.

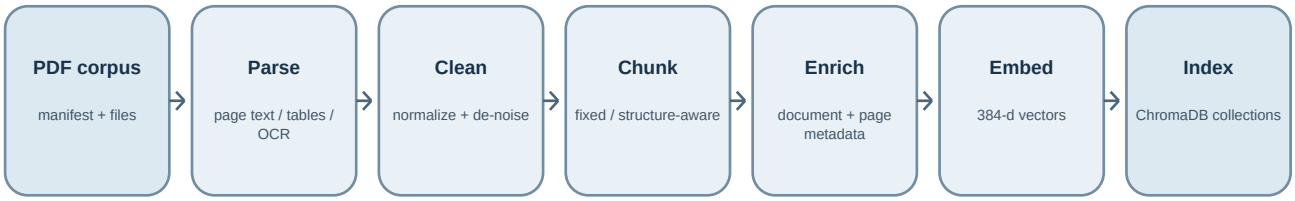
Scope. This report documents the architecture, implementation choices, retrieval evaluation, parser limitations, operational behaviour, and design considerations for scale and privacy. It is limited to the report deliverable; source code, README, manifest, evaluation files, logs, index evidence, and demo evidence remain separate deliverables.

Corpus snapshot. 37 PDF documents are represented in the manifest. The corpus contains 4,670 pages in the recorded full ingestion run, including 216 pages routed through OCR. Corpus composition is approximately 67.6% text-native-only, 18.9% table-heavy, and 13.5% scanned/mixed, so it is close to—but not exactly—the assignment's suggested 60/20/20 mix.

> **Important interpretation:** PolicyLens is a retrieval system, not a compliance decision engine. Retrieved passages must be checked against the cited source PDF, particularly for regulatory tables, numeric thresholds, and OCR-derived text.

1. Architecture and ingestion design

The pipeline separates document parsing, normalization, chunk construction, metadata enrichment, embedding, and vector indexing. Page-level provenance is retained so that a search result can be traced back to its source document and page range.



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Parsing and OCR

PyMuPDF is used for native PDF text and page-level extraction. OCR triage identifies pages with very low extracted text density or substantial image coverage; the documented rule routes pages with fewer than 20 words, or pages with at least 50% image coverage and fewer than 400 words, to Tesseract OCR. OCR pages are rendered at 2× before recognition. This hybrid approach avoids applying OCR to every text-native page while providing a fallback for image-based pages.

Cleaning

The cleaning stage normalizes whitespace, repairs line-break hyphenation, and removes detected repeating headers/footers and page-number noise. Native extracted text is retained for inspection. The manual cleaning example on a SEBI depositories circular page shows whitespace normalization while retaining the risk labels and values in the cleaned text. It also exposes a separate table-reconstruction defect: the reconstructed table rendered 1.5 as “15 .”. That defect is not attributed to whitespace cleaning.

Chunking comparison

Two strategies are evaluated. Fixed-size chunking uses a 500-token target with 50-token overlap. Structure-aware chunking follows headings and clause boundaries, with a documented target around 400 tokens and bounds of 300–500 tokens. Both strategies use the same corpus, embedding model, and retrieval evaluation protocol. Keeping the remaining settings constant makes the comparison attributable primarily to chunk construction.

Metadata and versioning

Each indexed passage carries document identity and provenance fields: doc_id, title, issuer, issue_date, page_start, page_end, section_title, and source_hash. These fields support issuer filtering, source attribution, page verification, and source-version replacement. A source hash provides a stable fingerprint for detecting unchanged PDFs and identifying revised files.

2. Embedding, indexing, and operational controls

Embedding model

The implementation uses the local Sentence Transformers model sentence-transformers/multi-qa-MiniLM-L6-cos-v1, producing 384-dimensional embeddings. Vectors are normalized and generated in batches of 32. The documented retry policy allows up to three retries for transient embedding failures. Because embeddings are computed locally, the recorded hosted embedding API cost is ■0; this does not include the compute, memory, or electricity cost of local execution.

Vector index

ChromaDB stores the vectors and their metadata, using cosine similarity. Separate collections are maintained for fixed-size and structure-aware chunks so that retrieval can be compared without mixing chunking strategies. The search interface returns rank, similarity score, document title, page range, section title, and a short chunk preview. The CLI supports repeatable metadata filters, including issuer; the current Streamlit UI does not expose the same filter controls.

Idempotency and replacement

Ingestion computes a source hash and checks the indexed source version before writing. If both strategy collections already contain the same source hash, the document is skipped. For a changed source, the new version is ingested before stale source records are removed, reducing the risk of leaving the document unavailable during replacement. Operational logs record ingested, skipped, missing, and failed documents.

Recorded operational evidence

Measure	Recorded value	Interpretation
Full corpus ingestion	4,670 pages; 897.46 s (14m57s)	Recorded full run; 37-document corpus snapshot
Full-run chunk counts	5,320 fixed; 12,920 structure-aware	Ingestion-run counters; not evaluation DB counts
Unchanged rerun	37 skipped; 0 ingested; 0 failed; 0 missing; 1.92 s	Evidence of hash-based idempotency
Evaluation collections	5,973 fixed; 14,157 structure-aware	Separate later evaluation database snapshot
Index footprint	259 MB total; fixed vectors 9.9 MB; structure vectors 25 MB	Shared SQLite file is not reliably attributable per collection

Latency caveat. Query latency is measured for the Chroma search call only; it excludes model loading and query embedding. The benchmark record reports fixed-size median 2.318 ms / p95 4.987 ms and structure-aware median 2.352 ms / p95 2.893 ms. These are local benchmark measurements, not end-to-end user-perceived latency.

3. Retrieval evaluation and findings

The evaluation set contains 25 realistic compliance questions, including five table-value questions and five materially paraphrased questions. Each question has a gold document and inclusive gold page range. A retrieval hit is counted when a result belongs to the gold document and overlaps the gold page range. Metrics are Recall@3, Recall@5, Recall@10, and mean reciprocal rank (MRR). Both strategies were evaluated over the same questions with top_k=10.

Overall and subset metrics

Subset	Strategy	N	R@3	R@5	R@10	MRR
Overall	Fixed	25	52%	56%	76%	0.5278
Overall	Structure-aware	25	60%	72%	80%	0.5787
Table questions	Fixed	5	40%	40%	80%	0.4536
Table questions	Structure-aware	5	80%	80%	80%	0.8000
Paraphrased	Fixed	5	60%	60%	100%	0.6536
Paraphrased	Structure-aware	5	60%	60%	80%	0.5333

Structure-aware chunking performs better overall: Recall@5 rises from 56% to 72%, Recall@10 from 76% to 80%, and MRR from 0.5278 to 0.5787. Its strongest advantage is on the table-question subset, where Recall@3 and Recall@5 are both 80%. It is not universally better: fixed-size chunking performs better on the paraphrased subset at Recall@10 (100% vs 80%) and MRR (0.6536 vs 0.5333). Structure-aware indexing also stores about 2.37 times as many chunks in the evaluation snapshot (14,157 vs 5,973), which increases index footprint.

Three disagreement examples

q04: Structure-aware retrieved the gold page (page 38) at rank 1; fixed-size did not retrieve a gold-page hit in its top 10. This is a clear structure-aware win.

q17: Structure-aware retrieved a gold-page result at rank 1 and a table passage at rank 3; fixed-size's first gold hit was rank 8. This favours structure-aware ranking, but the exact table-cell/column association still requires visual confirmation.

q13: Fixed-size retrieved the gold page (pages 69–71) at rank 5; structure-aware had no gold-page hit in its top 10. This is a clear fixed-size win and shows that structural chunking can miss relevant material.

A fourth reviewed example, q14, improved from fixed-size rank 7 to structure-aware rank 2. The rank-2 preview did not expose the PEP controls, so this is evidence of improved ranking only—not proof that the required control detail was retrieved.

4. Parsing failure analysis and quality limits

A manual inspection sampled 10 pages (seed 2026) from a 5,050-page PDF frame. All 10 sampled pages were text-native; none triggered OCR. Five pages contained detected tables. The sample therefore provides evidence about native text extraction and selected table cases, but it does not establish OCR accuracy or scanned-page quality.

Cleaning and table observations

Whitespace and values: In the SEBI depositories circular example (PDF page 170), native and cleaned text retain the risk labels and values, including 1.5. The separate reconstructed table output renders the value as “15 .”, indicating a table reconstruction/serialization problem rather than a cleaning-stage value change.

Wide, multi-level tables: One inspected table contains fragmented headers and multi-level column labels. Flattening the table into text makes the relationship between a value and its row/column heading difficult to recover. This is especially risky for threshold, percentage, and eligibility questions.

Cross-reference loss: Another inspected table preserves the text “Principle 6, Question 7 of Essential Indicators” in native extraction and reconstructed table output, but the cleaning preview omits that cross-reference field. This is a table-cell content preservation issue.

Wrapped rows: A further table contains wrapped circular numbers, dates, and subjects. The extracted layout makes row associations harder to assess; the sample does not independently prove that a particular row was incorrectly associated.

What was done and what remains

The pipeline retains page-level provenance and section metadata, supports table-aware chunking, and records extracted content for inspection. The evaluation includes five table questions, and the manual inspection explicitly records table reconstruction and cell-preservation risks. These controls improve traceability, but they do not eliminate layout ambiguity.

The following checks remain limitations rather than completed validation: the 10-page sample did not include OCR pages; OCR word accuracy was not measured; the inspected table pages were not visually verified against rendered source pages; and the sample is too small to establish corpus-wide parser accuracy. The 37-document corpus also has incomplete table/layout annotation coverage, so category proportions should be interpreted as inventory labels rather than a fully audited layout census.

> **Operational recommendation.** For high-impact regulatory values, verify the retrieved passage against the original PDF page. Preserve native text alongside serialized table output, and add a separate OCR evaluation set with manually transcribed ground truth before making accuracy claims about scanned documents.

The manual inspection supports a bounded conclusion: whitespace normalization generally preserved sampled native text, while table serialization and cell preservation produced identifiable risks. It does not support a claim of perfect parsing, perfect OCR, or complete visual validation.

5. Scaling, circular lifecycle, and privacy

Scaling to 100,000 documents

At 100,000 documents, the current single-machine design would need to evolve into a partitioned, observable ingestion service. A production design should use a durable object store for source PDFs, a metadata/catalogue database for document state and permissions, a queue for asynchronous ingestion, and horizontally scalable workers for parsing, OCR, and embedding. Batch sizes, concurrency, retry budgets, and back-pressure should be tuned against measured throughput and provider limits. The vector index should be sharded or partitioned by tenant and/or business domain, with metadata filters applied before or during retrieval. Operational dashboards should track ingestion lag, parse/OCR failures, index freshness, query latency, and storage growth.

Superseded and withdrawn circulars

A document should be represented as a versioned source, not overwritten without history. Store a stable document identity, source hash, effective/issue dates, version status (active, superseded, withdrawn), and a link to the replacing circular where available. On update, ingest and validate the new version, then atomically switch the active-version pointer and retire stale chunks from the active retrieval view. Retain prior versions for audit and historical queries, but exclude them from default current-policy search unless the user explicitly requests historical material. Withdrawn sources should be clearly labelled and excluded from ordinary active-policy results.

Access control and data privacy

Internal policies may contain confidential procedures, customer-related examples, control weaknesses, or commercially sensitive information. A shared index must not imply shared access. Enforce tenant- and role-based authorization at document ingestion and query time; apply access filters before results are returned; prevent cross-tenant retrieval; and ensure cached results, logs, exports, and backups follow the same access policy. Encrypt source files, metadata, embeddings, and backups at rest and in transit. Keep secrets out of source control, redact personal data from logs, define retention/deletion rules, and maintain audit trails for document access and administrative changes. Embeddings should be treated as sensitive derived data, not automatically as anonymous data.

Conclusion

The evaluated evidence favours structure-aware chunking overall and especially for the table-question subset, while fixed-size chunking remains stronger for the paraphrased subset. The pipeline demonstrates page-linked retrieval, local embeddings, hash-based unchanged-run skipping, and repeatable evaluation. Its principal quality risk is not basic whitespace cleaning but loss or ambiguity in complex table structure; OCR quality and visual source alignment remain unmeasured. PolicyLens is therefore a useful retrieval foundation, provided that high-impact results are checked against source pages and future production deployment adds stronger table/OCR validation, version lifecycle controls, and document-level authorization.

Evidence references within the project: `eval/rebenchmark_2026-11-07/metrics_phase3.csv`, `retrieval_log_phase3.csv`, `selected_disagreement_evidence.csv`, `q14_evidence.csv`, `run_metadata_phase3.json`; `eval/benchmark_run_2026-11-07.md`; `eval/manual_inspection_findings.md`; `eval/manual_inspection_run1/inspection_report.md`; `eval/gold_set_review.md`; `docs/ingestion_design.md`; `data/manifest.csv`.

Validation note: Gold-set validation passed for 25 questions (5 table, 5 paraphrased), unique ordered IDs, and positive inclusive page ranges within manifest bounds. Automated test suite: 21 passed. These checks validate structure and tested behaviour; they do not establish semantic correctness of every gold-page label or visual correctness of all extracted PDF content.